

01 / Three harnesses. Four box wires.

POOLDose

REV 14

Ferrule branches at existing devices · all 120 V wires are 14 AWG · small AC terminals each receive one blue ferrule

A AC supply / B Switching / C Low voltage · device order matches the enclosure photo

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification

Ferrules and checkout

Three LARGE RED ferrules: F1 LOAD, K5-3 and K1-3. One YELLOW: K5-2. Remaining 14 AWG device ends: BLUE.

Check: supply output 5.00 V with all relays closed; communication, timed run, STOP and flow-loss shutdown.

Calibrate with water before chemical service (docs/pump-calibration.md). Assembly and checkout: docs/build-instructions.html.

WARNING: K5-3 and K1-3 exceed the Finder terminal rating. Reference build only; do not copy. See docs/wiring-schedule.md.

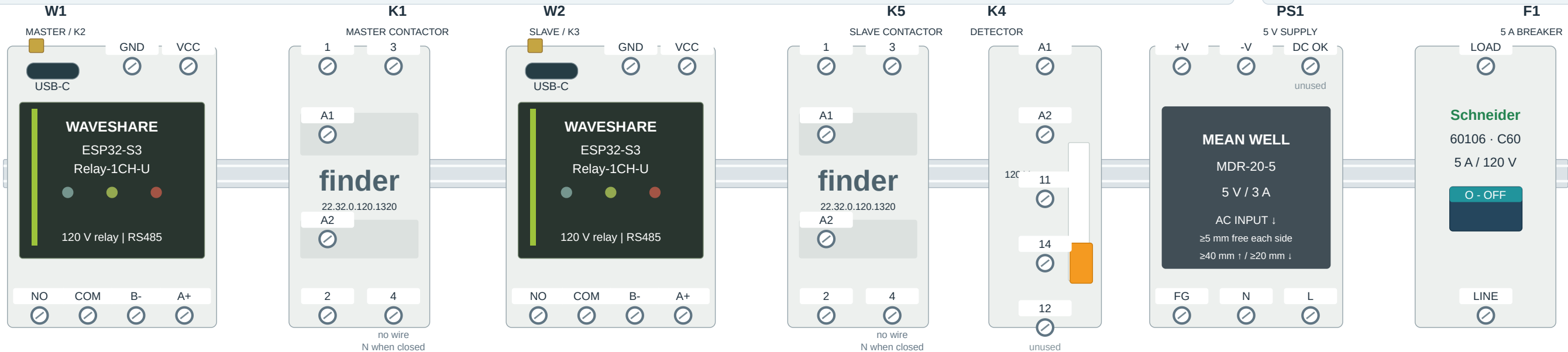
Finder operating power

A1/A2: 120 V AC coil supply.

1-2 and 3-4: NO contact pairs.

LED + mechanical indicator.

Wiring scope: see end notes.



A — AC SUPPLY

Black hot, white neutral, green PS1 ground.

Larger screws carry the branch ferrules.

Sheets 02–03.

B — SWITCHING

Red coil leads + outlet return.

K5-2 shares B04 + B05.

Sheet 04.

C — LOW VOLTAGE

5 V, RS485 + GPIO tails.

PS1 → W2 → W1 power chains.

Sheets 05–07.

Only these leads cross the metal-box boundary

A01 BLACK / 14 AWG OUT → F1 bottom LINE. A02 leaves the top LOAD terminal.

A08 WHITE / 14 AWG OUT → K5 POWER terminal 3. Both whip neutrals join inside the box.

B04 RED / 14 AWG IN ← final switched output. Wago → outlet-whip BLACK.

A17 GREEN / 14 AWG OUT → Mean Well PS1 FG. Ground Wago with both whip grounds.

Two 3-wire whips end in the box; all whip and 120 V harness wires are stranded 14 AWG.

Whip grounds are joined and bonded to the metal box.

METAL JUNCTION BOX

Wagos join the source/outlet whips to four control leads.

A01 · BLACK OUT → F1 bottom LINE

A08 · WHITE OUT → controls

B04 · RED IN → outlet-whip BLACK

A17 · GREEN OUT → PS1 FG

Neutral → outlet whip

Ground → outlet whip

PE → metal-box bond

Neutral and PE are separate nets. Green A17 is the only harness PE lead: ground Wago → PS1 FG.

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

6 requested shared ferrules

02 / Harness A / BLACK hot branches + GREEN ground

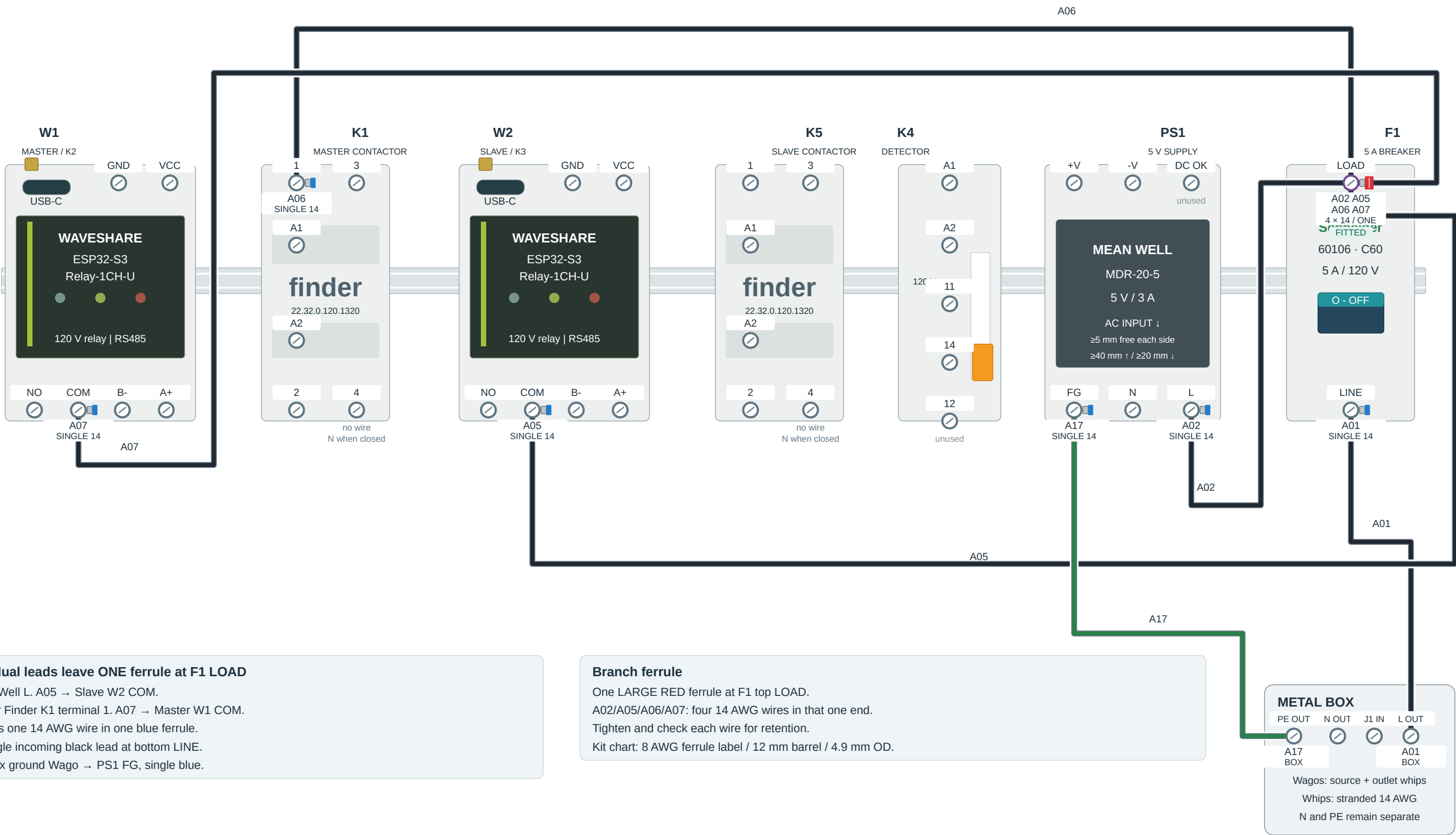
POOLDose

REV 14

F1 top LOAD: one ferrule / four 14 AWG wires → PS1 L, W2 COM, K1 terminal 1 and W1 COM · A17 box ground → PS1 FG

BLACK = stranded 14 AWG · GREEN A17 = PE · every destination is SINGLE BLUE · Waveshare COM is the relay contact, never VCC

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification



WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

This sheet: 1 shared ferrule

03 / Harness A / WHITE neutral branches

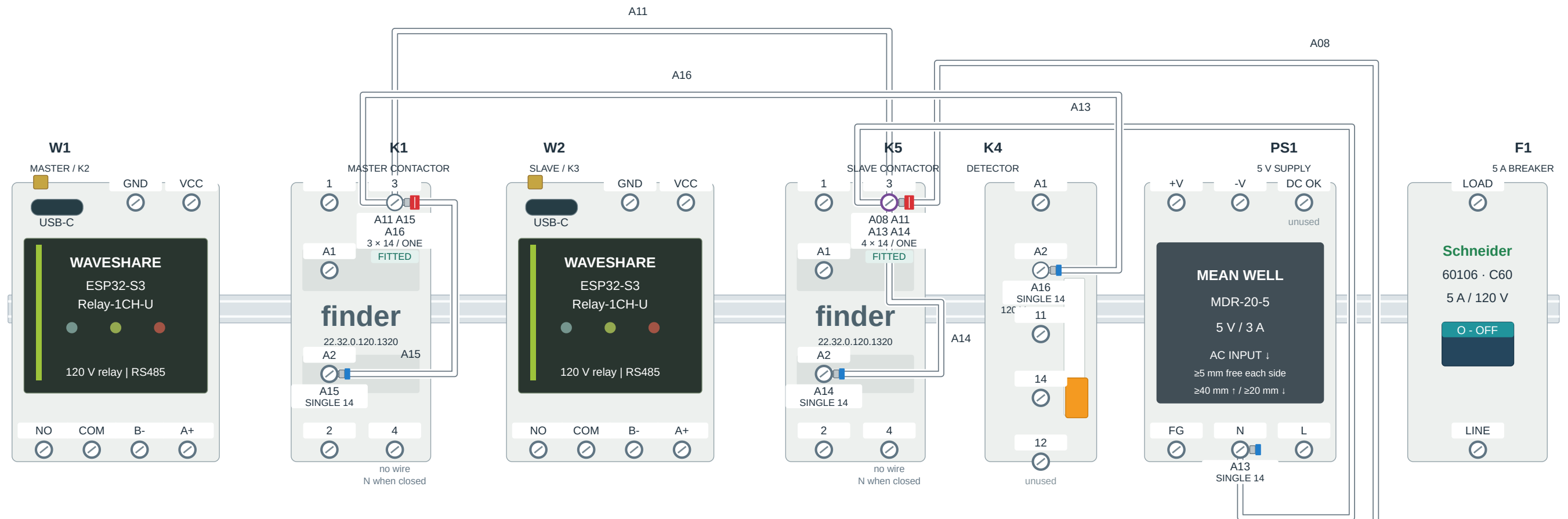
BOX N → Finder K5 terminal 3 → Finder K1 terminal 3 · separate single-blue leads to all four neutral loads

POOLDose

REV 14

WHITE = stranded 14 AWG · branch at POWER terminal 3 · all A1/A2 and Mean Well N ends are SINGLE BLUE

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification



K1 POWER terminal 3 / three wires

A11 IN + A15 to K1 A2 + A16 to K4 A2.
Over Finder rating: reference build only.
K1 A2 gets only A15 in one blue ferrule.

K5 POWER terminal 3 / four wires

A08 IN + A11 bridge + A13 + A14.
A13 → PS1 N. A14 → K5 A2; each single blue.
Over Finder rating: reference build only.

No neutral path crosses a contact

A11 is a real wire between the two terminal 3s.
Both terminal 4s have no external wire.
A closed 3–4 contact only makes its empty 4 neutral.

METAL BOX

N OUT J1 IN L OUT

A08 BOX

Wagos: source + outlet whips
Whips: stranded 14 AWG
N and PE remain separate

A1/A2 are coil terminals, distinct from 1/2/3/4. Both coils retain neutral in every contactor state.

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

This sheet: 2 shared ferrules

04 / Harness B / switched wires

POOLDose

REV 14

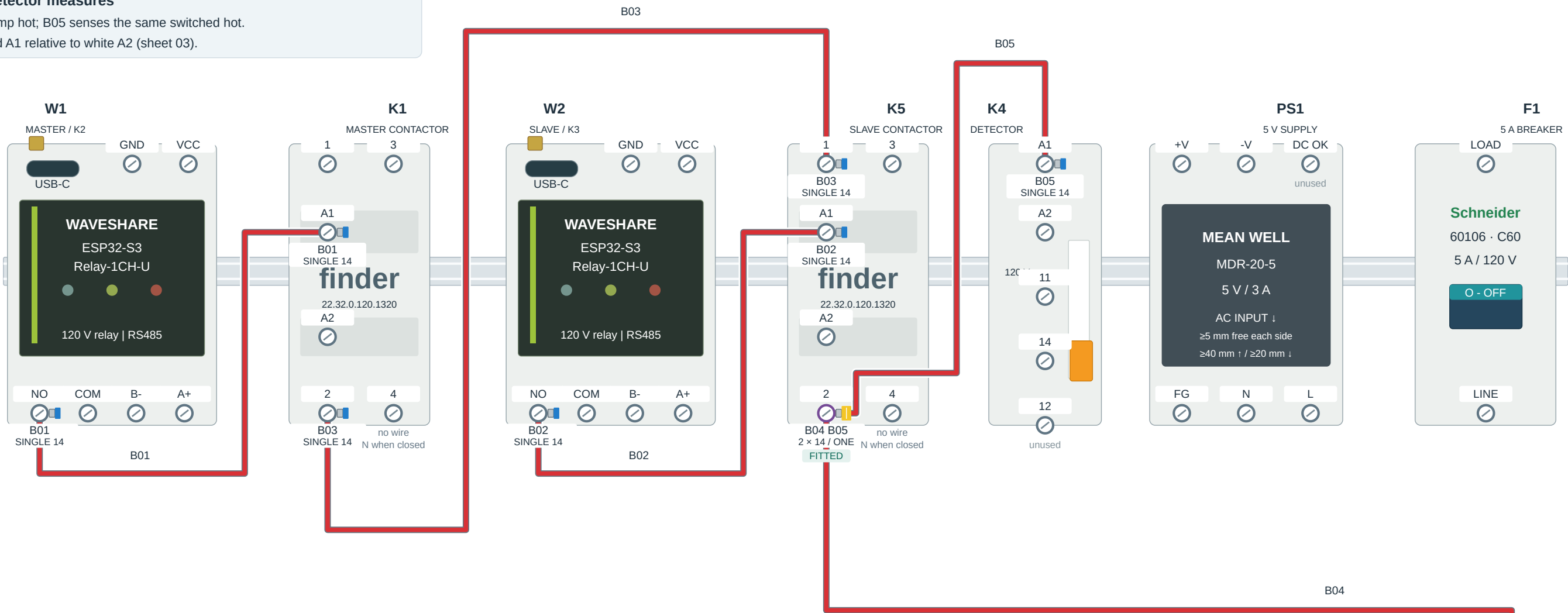
B01–B05 · all RED 14 AWG · independent coil jumpers and series pump-power path

B01 = Master → K1 coil · B02 = Slave → K5 coil · B03 = series power link · B04/B05 = final output + detector

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification

What the detector measures

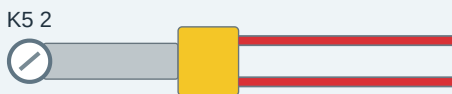
B04 feeds pump hot; B05 senses the same switched hot.
K4 senses red A1 relative to white A2 (sheet 03).



Finder coil power: A1/A2 are distinct from 1/2

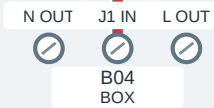
B01: W1 NO → K1 A1. B02: W2 NO → K5 A1.
Both A2 terminals receive white neutral from harness A.
1-2 switches pump hot. Terminal 3 is neutral; 4 has no wire.
The LED follows coil power; indicators need no extra wiring.

Both red wires share ONE ferrule at Finder K5 terminal 2



Detector A1 receives B05 alone: SINGLE 14.
Its A1 and A2 each receive one 14 AWG wire in one ferrule.
K5-2: one YELLOW ferrule holds both red 14 AWG wires.

METAL BOX



Wagos: source + outlet whips
Whips: stranded 14 AWG
N and PE remain separate

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

This sheet: 1 shared ferrule

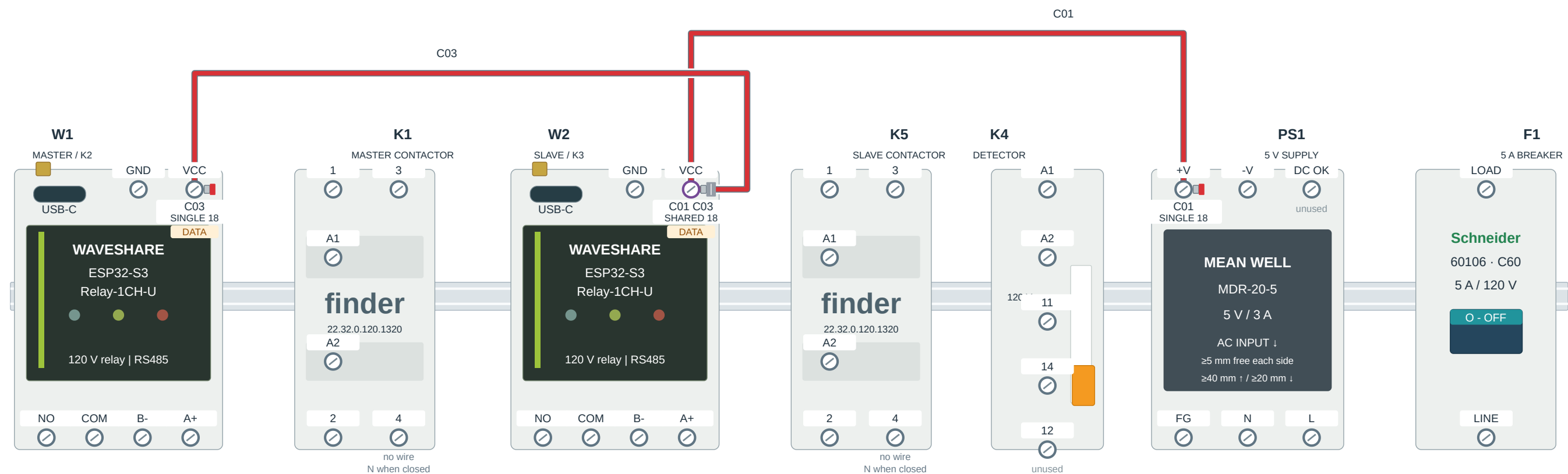
05 / Harness C / RED +5 V daisy chain

POOLDose

REV 14

5 V DC ONLY · C01 arrives and C03 continues from the SAME W2 VCC screw

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification



The red low-voltage harness

C01: PS1 +V → W2 VCC.

C03: same W2 VCC → W1 VCC.

W2 VCC: C01 + C03 together in ONE 2 × 18 AWG twin.

PS1 +V and W1 VCC each have one single-ferruled wire.

Existing terminal checks

DATA: record Waveshare single/twin wire ranges, strip and torque.

PS1 stays single on +V; DC OK stays unused.

Use the top VCC screw, not GPIO 3V3.

The requested shared end is visible at W2; no added connector is drawn.

06 / Harness C / BLACK 0 V daisy chain + RS485

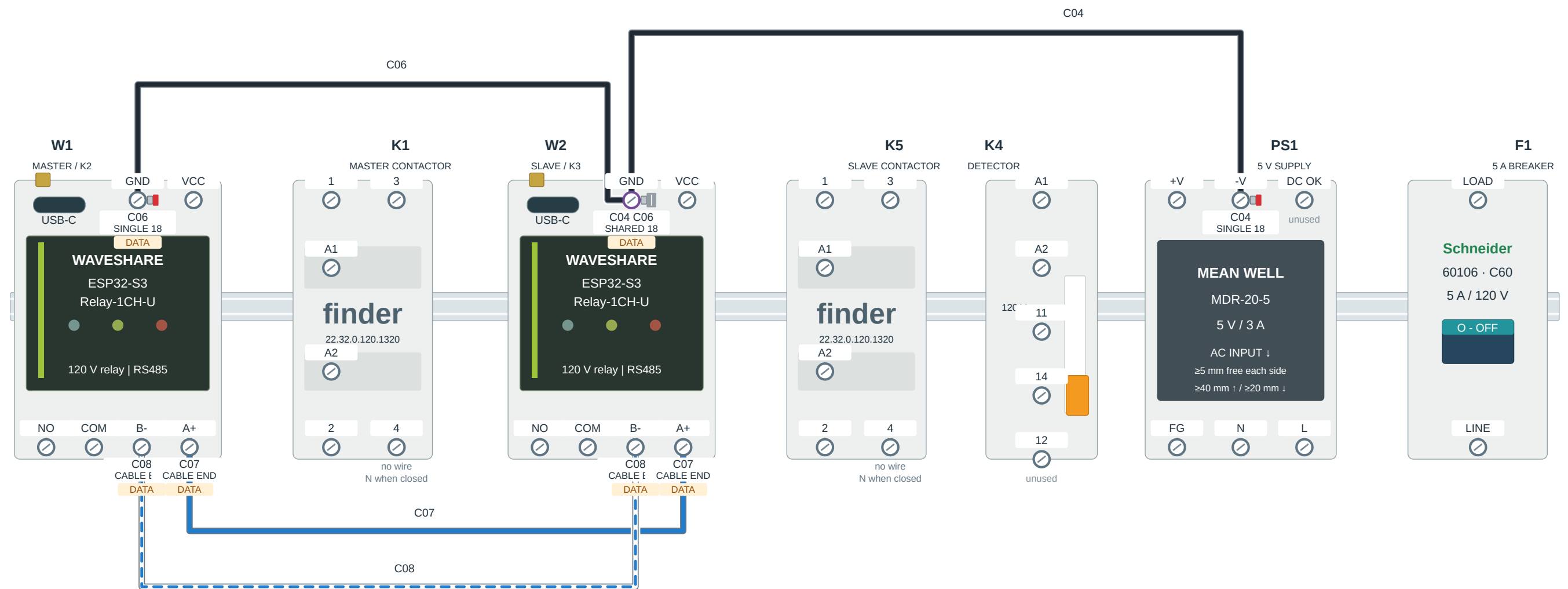
Two BLACK 18 AWG segments · PS1 -V → Slave GND → Master GND

POOLDose

REV 14

BLACK = DC 0 V, separate from AC neutral and PE · BLUE = A+ · WHITE/BLUE = B-

SINGLE = one conductor · SHARED = listed conductors in ONE ferrule · FITTED = as built · DATA = remaining specification



Intact Cat6A cable

C07 blue: A+ ↔ A+. C08 white/blue: B- ↔ B-.

Preserve the cable jacket and pair twist.

Enable both 120 Ω jumpers.

Check cable gauge/construction; solid conductors stay bare.

The black low-voltage harness

C04: PS1 -V → W2 GND. C06: same W2 GND → W1 GND.

W2 GND: C04 + C06 together in ONE 2 × 18 AWG twin.

PS1 -V and final W1 GND each have one single-ferruled wire.

DATA: record the Waveshare terminal/ferrule specification.

07 / Harness C / GPIO plugs and pull-up

POOLDose

REV 14

Photographed local pull-up splice · ORANGE detector signal / BLACK return · completed RED/GREEN telephone-cord flow tail

J5 plug colors: 1 YELLOW / GPIO1 · 2 ORANGE / GPIO2 · 3 RED / +3.3 V · 4 BLACK / GND

Slave: RED → R1 → one splice with YELLOW GPIO1 + ORANGE C13. BLACK → BLACK C14. Insulate each joint separately.

C13 alone → detector 14; C14 alone → detector 11. Both are single-wire ends. The thin ORANGE GPIO2 lead stays unused.

W2 / SLAVE · four-wire J5 plug

Photo layout · orange signal / black return

1 · GPIO1 · YELLOW

2 · GPIO2 · ORANGE

3 · +3.3 V · RED

4 · GND · BLACK

YELLOW GPIO1 joins the resistor/output splice at JS2

Cap this ORANGE plug wire individually. GPIO2 has no connection.

R1 · 270 Ω · ¼ W

RED lead joined directly to R1
Either resistor lead may face 3.3 V.
C12 is the resistor's own lead; sleeve it.

JS2: YELLOW GPIO1 + R1 lead C12 + ORANGE C13 in one local splice.

K4-14 and K4-11 each receive ONE 18 AWG wire in a SINGLE ferrule.

Insulate JS2, the RED/R1 splice and the BLACK splice separately; support the assembly.
C14 · BLACK · to K4-11

K4 / DETECTOR

Contact terminals only

11
C14
SINGLE 18

11 = common
14 = normally open

11–14 closes when
the outlet is powered.

14
C13
SINGLE 18

12
unused

Unplug GPIO plugs for soldering and resistance checks. Slave pins 3 ↔ 1: 270 Ω ±5%; pins 1 ↔ 4: open with K4 released.

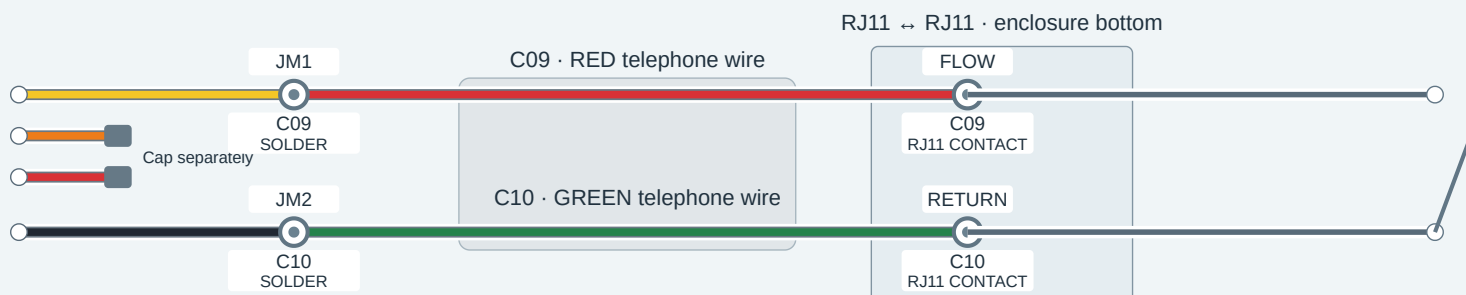
W1 / MASTER · completed flow tail

1 · GPIO1 · YELLOW

2 · GPIO2 · ORANGE

3 · +3.3 V · RED

4 · GND · BLACK



Photos: YELLOW GPIO1 → RED cord wire; BLACK GND → GREEN cord wire. Master tail is shown finished in heat-shrink.

S1 · generic Hayward-style flow switch

RJ11 plug connects to the other side.

Keep the installed contact mapping.

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

This sheet: 0 shared ferrules

08 / One ferrule / multiple conductors

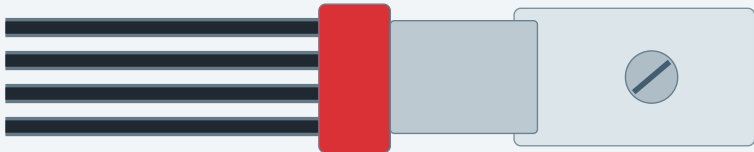
Six shared device locations · K4 detector terminals 11 and 14 now each receive a SINGLE 18 AWG wire

POOLDose

REV 14

F1 top LOAD / 4 × 14 AWG

A02 + A05 + A06 + A07



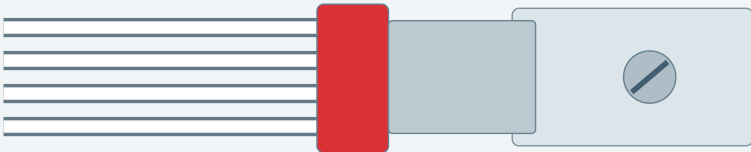
ONE crimped end / 4 separate conductors

LARGE RED / 12 mm kit barrel.

Single blue at each small-terminal destination.

K5 POWER terminal 3 / 4 × 14 AWG

A08 + A11 + A13 + A14



ONE crimped end / 4 separate conductors

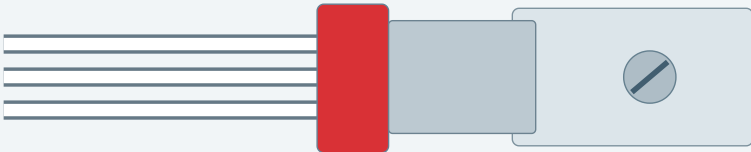
LARGE RED / 12 mm kit barrel.

Single blue at each small-terminal destination.

Exceeds Finder rating: reference build only.

K1 POWER terminal 3 / 3 × 14 AWG

A11 + A15 + A16



ONE crimped end / 3 separate conductors

LARGE RED / 12 mm kit barrel.

Single blue at each small-terminal destination.

Exceeds Finder rating: reference build only.

EXISTING SCREW	ALL WIRES IN ONE FERRULE	INSULATION / AWG	TERMINAL NOTE
F1 LOAD	A02 + A05 + A06 + A07	BLACK · 4 × 14	FITTED — LARGE RED; tightened and connected as drawn.
K5 terminal 3	A08 + A11 + A13 + A14	WHITE · 4 × 14	FITTED — LARGE RED; exceeds Finder rating, reference build only.
K1 terminal 3	A11 + A15 + A16	WHITE · 3 × 14	FITTED — LARGE RED; exceeds Finder rating, reference build only.
K5 terminal 2	B04 + B05	RED · 2 × 14	FITTED — YELLOW; two 14 AWG wires in the one kit ferrule.
W2 VCC	C01 + C03	RED · 2 × 18	DATA — record twin size/profile within the blue-sized envelope.
W2 GND	C04 + C06	BLACK · 2 × 18	DATA — record twin size/profile within the blue-sized envelope.

14 AWG ferrules: three large red, one yellow, remaining device ends blue

Mean Well L/N/FG, Waveshare COM/NO, detector A1/A2 and all Finder A1/A2: one 14 AWG wire in one blue ferrule.

14 AWG groups: two four-wire red, one three-wire red, one two-wire yellow and 17 single blue ends.

Kit chart: large red = 8 AWG label / 12 mm / 4.9 mm OD; yellow = 10 AWG label / 12 mm / 3.9 mm OD.

K4-14: one orange C13. K4-11: one black C14. The GPIO1/R1 branch is soldered locally at JS2; no shared ferrule at K4.

Finder 22.32 contact terminals are rated for 1 × 10 AWG or 2 × 12 AWG. K5-3 and K1-3 exceed that; use a rated neutral terminal block instead.

WIRE BY ID · follow the end-preparation notes and assembly checkout · crossings do not connect

6 requested shared ferrules

Every wire end

Harness revision 14 — ferrule branches at existing devices. 28 IDs / 56 ends; six shared ferrules. Set the Mean Well MDR-20-5 supply to 5 V. After assembly, measure 5.00 V at the DC supply output with all relays closed, then confirm Home Assistant/RS485 communication, outlet sensing, a normal timed run, STOP and flow-loss shutdown. Calibrate with water before chemical service; see pump-calibration.md. Completed wiring does not replace these measurements. SINGLE = one conductor; SHARED = the stated number of conductors in one ferrule. All 120 V wiring stays 14 AWG. FITTED = as built; DATA = specifications to record; CHECK = end preparation. C12 is an integral resistor lead.

Wire	Insulation / conductor	End A · exact location	A · prepare this end	End B · exact location	B · prepare this end
A01	Black 14 AWG MTW, 600 V	BOX:L Inside metal box: hot Wago joins source-whip black to separate black A01 lead OUT	BOX Wago connection inside the metal box; follow the installed connector conductor range and strip length. No ferrule specified at this end.	F1:LINE F1 bottom LINE clamp; A01 incoming supply	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. C60 wire guidance: 14 mm strip and 22 lb-in for this current/voltage range, subject to installed markings. Kit blue barrel is 8 mm; check the metal-barrel engagement against the device instructions.
A02	Black 14 AWG MTW, 600 V	F1:LOAD F1 top LOAD clamp; one installed large-red ferrule with four black 14 AWG branches	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A02/A05/A06/A07, four black 14 AWG wires, tightened and connected as drawn. Wires: A02 + A05 + A06 + A07. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	PS1:L PS1 bottom RIGHT L	SINGLE 14 Single blue 14 AWG ferrule. The design limits all Mean Well terminals to a blue-single-14-AWG ferrule envelope; confirm the selected ferrule/barrel. MDR manual: copper, at least 80°C insulation, 6.5 mm strip, 5.0 lb-in (0.57 N·m); kit barrel is 8 mm.
A05	Black 14 AWG MTW, 600 V	F1:LOAD F1 top LOAD clamp; one installed large-red ferrule with four black 14 AWG branches	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A02/A05/A06/A07, four black 14 AWG wires, tightened and connected as drawn. Wires: A02 + A05 + A06 + A07. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	W2:COM W2 Slave / K3: bottom 2nd screw COM	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
A06	Black 14 AWG MTW, 600 V	F1:LOAD F1 top LOAD clamp; one installed large-red ferrule with four black 14 AWG branches	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A02/A05/A06/A07, four black 14 AWG wires, tightened and connected as drawn. Wires: A02 + A05 + A06 + A07. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K1:1 K1 Finder: top outer LEFT screw 1	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm.
A07	Black 14 AWG MTW, 600 V	F1:LOAD F1 top LOAD clamp; one installed large-red ferrule with four black 14 AWG branches	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A02/A05/A06/A07, four black 14 AWG wires, tightened and connected as drawn. Wires: A02 + A05 + A06 + A07. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	W1:COM W1 Master / K2: bottom 2nd screw COM	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
A08	White 14 AWG MTW, 600 V	BOX:N Inside metal box: neutral Wago joins source-whip neutral, outlet-whip neutral and white A08 lead OUT	BOX Wago connection inside the metal box; follow the installed connector conductor range and strip length. No ferrule specified at this end.	K5:3 K5 Finder: top outer RIGHT / neutral branch screw 3	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A08/A11/A13/A14, four white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A08 + A11 + A13 + A14. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.

Wire	Insulation / conductor	End A · exact location	A · prepare this end	End B · exact location	B · prepare this end
A11	White 14 AWG MTW, 600 V	K5:3 K5 Finder: top outer RIGHT / neutral branch screw 3	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A08/A11/A13/A14, four white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A08 + A11 + A13 + A14. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K1:3 K1 Finder: top outer RIGHT / neutral branch screw 3	3 × 14 / ONE FITTED: one LARGE RED ferrule holding A11/A15/A16, three white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A11 + A15 + A16. ONE LARGE RED ferrule containing THREE 14 AWG conductors, at K1 terminal 3. Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.
A13	White 14 AWG MTW, 600 V	K5:3 K5 Finder: top outer RIGHT / neutral branch screw 3	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A08/A11/A13/A14, four white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A08 + A11 + A13 + A14. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	PS1:N PS1 bottom MIDDLE N	SINGLE 14 Single blue 14 AWG ferrule. The design limits all Mean Well terminals to a blue-single-14-AWG ferrule envelope; confirm the selected ferrule/barrel. MDR manual: copper, at least 80°C insulation, 6.5 mm strip, 5.0 lb-in (0.57 N·m); kit barrel is 8 mm.
A14	White 14 AWG MTW, 600 V	K5:3 K5 Finder: top outer RIGHT / neutral branch screw 3	4 × 14 / ONE FITTED: one LARGE RED ferrule holding A08/A11/A13/A14, four white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A08 + A11 + A13 + A14. ONE LARGE RED ferrule containing FOUR 14 AWG conductors, Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K5:A2 K5 Finder: lower inner ledge screw A2	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm. A1/A2: one 14 AWG wire in one blue ferrule; no branch at this coil terminal.
A15	White 14 AWG MTW, 600 V	K1:3 K1 Finder: top outer RIGHT / neutral branch screw 3	3 × 14 / ONE FITTED: one LARGE RED ferrule holding A11/A15/A16, three white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A11 + A15 + A16. ONE LARGE RED ferrule containing THREE 14 AWG conductors, at K1 terminal 3. Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K1:A2 K1 Finder: lower inner ledge screw A2	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm. A1/A2: one 14 AWG wire in one blue ferrule; no branch at this coil terminal.
A16	White 14 AWG MTW, 600 V	K1:3 K1 Finder: top outer RIGHT / neutral branch screw 3	3 × 14 / ONE FITTED: one LARGE RED ferrule holding A11/A15/A16, three white 14 AWG wires, tightened. Reference build only: exceeds the Finder terminal rating; see the known-limitation warning. Wires: A11 + A15 + A16. ONE LARGE RED ferrule containing THREE 14 AWG conductors, at K1 terminal 3. Pittsburgh 70195 kit chart: 8 AWG label, 12 mm barrel, 4.9 mm tube OD.	K4:A2 K4 top inner A2 input clamp	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. K4 supports single ferrules 0.2–2.5 mm²; strip 8 mm, torque 0.6–0.8 N·m.
A17	Green 14 AWG MTW, 600 V	BOX:G Inside metal box: ground Wago joins both whip grounds, the box bond and green A17 lead OUT	BOX Wago connection inside the metal box; follow the installed connector conductor range and strip length. No ferrule specified at this end.	PS1:FG PS1 bottom LEFT FG (⊥)	SINGLE 14 Single blue 14 AWG ferrule. The design limits all Mean Well terminals to a blue-single-14-AWG ferrule envelope; confirm the selected ferrule/barrel. MDR manual: copper, at least 80°C insulation, 6.5 mm strip, 5.0 lb-in (0.57 N·m); kit barrel is 8 mm.
B01	Red 14 AWG MTW, 600 V	W1:NO W1 Master / K2: bottom 1st / LEFTMOST screw NO	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.	K1:A1 K1 Finder: upper inner ledge screw A1	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm. A1/A2: one 14 AWG wire in one blue ferrule; no branch at this coil terminal.

Wire	Insulation / conductor	End A · exact location	A · prepare this end	End B · exact location	B · prepare this end
B02	Red 14 AWG MTW, 600 V	W2:NO W2 Slave / K3: bottom 1st / LEFTMOST screw NO	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.	K5:A1 K5 Finder: upper inner ledge screw A1	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm. A1/A2: one 14 AWG wire in one blue ferrule; no branch at this coil terminal.
B03	Red 14 AWG MTW, 600 V	K1:2 K1 Finder: bottom outer LEFT screw 2	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm.	K5:1 K5 Finder: top outer LEFT screw 1	SINGLE 14 Single blue 14 AWG ferrule. Catalog strip 9 mm, torque 0.8 N·m; kit barrel 8 mm.
B04	Red 14 AWG MTW, 600 V	K5:2 K5 Finder: bottom outer LEFT screw 2	2 × 14 / ONE FITTED: one YELLOW ferrule holding B04/B05, two red 14 AWG wires, tightened and connected as drawn. Wires: B04 + B05. ONE YELLOW ferrule containing TWO 14 AWG conductors, at K5 terminal 2. Pittsburgh 70195 kit chart: 10 AWG label, 12 mm barrel, 3.9 mm tube OD. This records the kit ferrule used, not a separate twin-ferrule product.	BOX:SW Inside metal box: switched-hot Wago joins red B04 lead IN to outlet-whip black	BOX Wago connection inside the metal box; follow the installed connector conductor range and strip length. No ferrule specified at this end.
B05	Red 14 AWG MTW, 600 V	K5:2 K5 Finder: bottom outer LEFT screw 2	2 × 14 / ONE FITTED: one YELLOW ferrule holding B04/B05, two red 14 AWG wires, tightened and connected as drawn. Wires: B04 + B05. ONE YELLOW ferrule containing TWO 14 AWG conductors, at K5 terminal 2. Pittsburgh 70195 kit chart: 10 AWG label, 12 mm barrel, 3.9 mm tube OD. This records the kit ferrule used, not a separate twin-ferrule product.	K4:A1 K4 top outer A1 input clamp	SINGLE 14 SINGLE 14 AWG ferrule. Kit: blue collar, 8 mm barrel; kit chart metal-tube OD 2.6 mm. Check device-specific note. K4 supports single ferrules 0.2–2.5 mm²; strip 8 mm, torque 0.6–0.8 N·m.
C01	Red 18 AWG MTW, 600 V	PS1:+V PS1 top LEFT +V	SINGLE 18 Single 18 AWG ferrule. The design limits all Mean Well terminals to a blue-single-14-AWG ferrule envelope; confirm the selected ferrule/barrel. MDR manual: copper, at least 80°C insulation, 6.5 mm strip, 5.0 lb-in (0.57 N·m); kit barrel is 8 mm.	W2:VCC W2 Slave / K3: top power RIGHT (+5 V) screw VCC	SHARED 18 DATA: record the two-18-AWG twin size/profile within the blue-sized envelope and the Waveshare clamp strip/torque specification. Wires: C01 + C03. ONE two-wire ferrule containing TWO 18 AWG conductors. Record its size, barrel length and crimp profile. Check retention of each wire and metal-barrel engagement at the clamp.
C03	Red 18 AWG MTW, 600 V	W2:VCC W2 Slave / K3: top power RIGHT (+5 V) screw VCC	SHARED 18 DATA: record the two-18-AWG twin size/profile within the blue-sized envelope and the Waveshare clamp strip/torque specification. Wires: C01 + C03. ONE two-wire ferrule containing TWO 18 AWG conductors. Record its size, barrel length and crimp profile. Check retention of each wire and metal-barrel engagement at the clamp.	W1:VCC W1 Master / K2: top power RIGHT (+5 V) screw VCC	SINGLE 18 SINGLE 18 AWG ferrule. Kit reference: SMALL red collar, 8 mm barrel; kit chart metal-tube OD 1.7 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
C04	Black 18 AWG MTW, 600 V	PS1:-V PS1 top MIDDLE −V	SINGLE 18 Single 18 AWG ferrule. The design limits all Mean Well terminals to a blue-single-14-AWG ferrule envelope; confirm the selected ferrule/barrel. MDR manual: copper, at least 80°C insulation, 6.5 mm strip, 5.0 lb-in (0.57 N·m); kit barrel is 8 mm.	W2:GND W2 Slave / K3: top power LEFT screw GND	SHARED 18 DATA: record the two-18-AWG twin size/profile within the blue-sized envelope and the Waveshare clamp strip/torque specification. Wires: C04 + C06. ONE two-wire ferrule containing TWO 18 AWG conductors. Record its size, barrel length and crimp profile. Check retention of each wire and metal-barrel engagement at the clamp.

Wire	Insulation / conductor	End A · exact location	A · prepare this end	End B · exact location	B · prepare this end
C06	Black 18 AWG MTW, 600 V	W2:GND W2 Slave / K3: top power LEFT screw GND	SHARED 18 DATA: record the two-18-AWG twin size/profile within the blue-sized envelope and the Waveshare clamp strip/torque specification. Wires: C04 + C06. ONE two-wire ferrule containing TWO 18 AWG conductors. Record its size, barrel length and crimp profile. Check retention of each wire and metal-barrel engagement at the clamp.	W1:GND W1 Master / K2: top power LEFT screw GND	SINGLE 18 SINGLE 18 AWG ferrule. Kit reference: SMALL red collar, 8 mm barrel; kit chart metal-tube OD 1.7 mm. Check device-specific note. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
C07	Blue One conductor of intact Cat6A pair	W1:A+ W1 Master / K2: bottom 4th / RIGHTMOST screw A+	CABLE END Identify Cat6A gauge/construction and clamp specification. Solid: bare. Stranded: only an accepted matching ferrule. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.	W2:A+ W2 Slave / K3: bottom 4th / RIGHTMOST screw A+	CABLE END Identify Cat6A gauge/construction and clamp specification. Solid: bare. Stranded: only an accepted matching ferrule. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
C08	White/blue stripe Mate of same intact Cat6A pair	W1:B- W1 Master / K2: bottom 3rd screw B-	CABLE END Identify Cat6A gauge/construction and clamp specification. Solid: bare. Stranded: only an accepted matching ferrule. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.	W2:B- W2 Slave / K3: bottom 3rd screw B-	CABLE END Identify Cat6A gauge/construction and clamp specification. Solid: bare. Stranded: only an accepted matching ferrule. DATA: record the installed Waveshare clamp wire/ferrule range, strip and torque.
C09	Red Installed telephone-cord conductor; gauge not recorded	JM1 JM1: Master J5-1 GPIO1 YELLOW factory lead to RED telephone-cord conductor C09; photographed splice	SOLDER Mechanically engaged solder joint, individually sleeved and strain-relieved; no ferrule.	RJ11:FLOW Installed RJ11 coupling at enclosure bottom: GPIO1 contact of the connected flow-switch pair; functional label, not a contact number	RJ11 CONTACT Existing telephone-cord RJ11 contact. Retain the installed contact mapping; this drawing assigns no numbered RJ11 pinout.
C10	Green Installed telephone-cord conductor; gauge not recorded	JM2 JM2: Master J5-4 GND BLACK factory lead to GREEN telephone-cord conductor C10; photographed splice	SOLDER Mechanically engaged solder joint, individually sleeved and strain-relieved; no ferrule.	RJ11:RETURN Installed RJ11 coupling at enclosure bottom: GND contact of the connected flow-switch pair; functional label, not a contact number	RJ11 CONTACT Existing telephone-cord RJ11 contact. Retain the installed contact mapping; this drawing assigns no numbered RJ11 pinout.
C12	Metal lead Integral R1 lead	R1:SENSE Integral R1 sense-side lead; C12 is not cut hookup wire	R1 LEAD Integral resistor lead C12; sleeve the exposed metal and its splice. No extra hookup wire or ferrule at this end.	JS2 JS2: ONE local three-conductor splice joins YELLOW GPIO1 factory lead, R1 lead C12 and ORANGE C13 to detector terminal 14	SOLDER Three-conductor solder splice: YELLOW GPIO1 factory lead + R1 lead C12 + ORANGE C13. Individually insulate this node from the RED 3V3 splice and support the resistor assembly.
C13	Orange 18 AWG MTW, 600 V	JS2 JS2: ONE local three-conductor splice joins YELLOW GPIO1 factory lead, R1 lead C12 and ORANGE C13 to detector terminal 14	SOLDER Three-conductor solder splice: YELLOW GPIO1 factory lead + R1 lead C12 + ORANGE C13. Individually insulate this node from the RED 3V3 splice and support the resistor assembly.	K4:14 K4 bottom middle 14 normally open	SINGLE 18 SINGLE 18 AWG ferrule. Kit reference: SMALL red collar, 8 mm barrel; kit chart metal-tube OD 1.7 mm. Check device-specific note. K4 supports single ferrules 0.2–2.5 mm²; strip 8 mm, torque 0.6–0.8 N·m.
C14	Black 18 AWG MTW, 600 V	JS3 JS3: Slave J5-4 GND BLACK factory lead joined to BLACK C14 for detector terminal 11	SOLDER Mechanically engaged solder joint, individually sleeved and strain-relieved; no ferrule.	K4:11 K4 bottom inner 11 common	SINGLE 18 SINGLE 18 AWG ferrule. Kit reference: SMALL red collar, 8 mm barrel; kit chart metal-tube OD 1.7 mm. Check device-specific note. K4 supports single ferrules 0.2–2.5 mm²; strip 8 mm, torque 0.6–0.8 N·m.